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Applied Corporate Finance Seminar

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Valuation of Rockwool A/S

A fundamental analysis with an emphasis on cost of capital

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Abstract

This paper presents a comprehensive valuation of Rockwool A/S, a leading manufacturer in the mineral wool sector, by employing a Discounted Cash Flow (DCF) model with a focus on the cost of capital. The paper finds a share price DKK 1,709.13 compared to the actual share price of DKK 1,976 as of December 31, 2023. The analysis integrates a deep dive into the financials of Rockwool, examining its profitability, market dynamics, and strategic positioning within the global insulation market. Emphasis is given to Rockwool's cost of capital, reflecting its financial structure and market risk, which is crucial for an accurate valuation. Despite challenges from geopolitical tensions and market volatility, Rockwool maintains a robust market presence, supported by sustainable practices and a strong commitment to energy efficiency and safety in building solutions. This paper also includes a multiple analysis that finds a share price much closer to the actual one, which leads to the conclusion that our primary valuation is associated with a high level of uncertainty.

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1 Introduction

In this paper you will be presented with an analysis that seeks to find the intrinsic value of the Danish company Rockwool A/S also known as ROCKWOOL Group. The valuation is based on the financial activity and economic environment at December 31, 2023.

The paper includes a deep dive into the financials of Rockwool and aims to value the company using a Discounted Cash-Flow model (DCF-model) with an emphasis on the cost of capital. The motivation for valuing Rockwool stems from their huge market role as a leading manufacturer of mineral wool as well as their controversial decision to keep operating in the war fighting Russia. The analysis will shed a light on the market dynamics for a public traded company in the construction sector and cross-check how the market values Rockwool with perspective for the future. The conclusive valuation will be of the date December 31, 2023 which limits the use in current times but with consideration for the economic development past the value date, the discussion and main points still holds useful insight to the fiscal value of the Rockwool.

2 The Company and Market Structure

2.1 Rockwool A/S

Rockwool A/S is a global leader in stone wool insulation products and solutions, headquartered in Hedehusene, Denmark. Founded in 1909, Rockwool has built a reputation on its commitment to enhancing sustainability and improving the safety and energy efficiency of buildings. The company's products are centered around a natural stone wool technology that provides thermal and acoustic insulation as well as fire resistance and durability. Rockwool operates in a diverse range of sectors, including residential, commercial, and industrial markets, and has a presence in over 40 countries, supported by approximately 12.000 employees worldwide.¹ This global footprint allows Rockwool to leverage local and international market dynamics to optimize production and distribution. Rockwool has been listed on the Copenhagen Stock Exchange since 1996 and is primarily owned by Danish investors with the biggest shareholder being Rockwool fonden with approximately 16% of the issued shares.²

2.2 Stock Performance

Rockwool's stock has consistently outperformed the MSCI World Index since 2015, achieving an annualized return of 12.2% from 2015 to 2023, and 10.5% from 2000 to 2023 as shown in Figure 1. Despite steady growth in the early 21st century, the stock experienced significant price volatility in recent years. Trade tensions between the USA and China in late 2018, particularly impacting the construction sector, led to a sharp decrease in earnings per share for Rockwool.³

Post-2018, the stock maintained steady financial performance, though earnings per share did not recover to previous levels, leading to reduced investor expectations and a significant price drop following the early 2020 market crash due to COVID-19. However, Rockwool's conservative financial approach during the pandemic allowed it to remain profitable and continue dividend payouts.⁴

The stock reached an all-time high of 3,342 DKK in 2021 but faced a downturn amid escalating conflicts in Ukraine and Russia, compounded by operational challenges in Russia. As of December

¹Rockwool website[1]

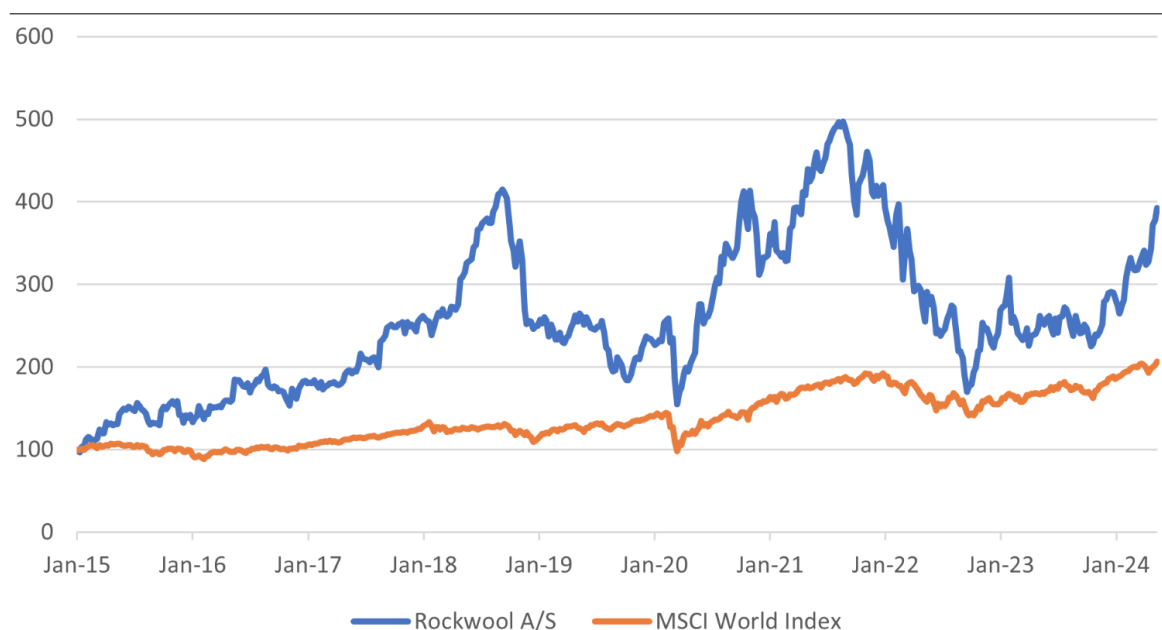
²Annual report 2023[7]

³The street (2019)[31]

⁴Sustainability report 2021[2]

31, 2023, the stock price stood at 1,976 DKK. This backdrop sets the stage for a deeper strategic analysis of Rockwool's future prospects amid ongoing market uncertainties.⁵

Figure 1. Rockwool A/S Stock Performance



Source: Yahoo Finance

Note: Index prices are normalized to January 2015 and are based on adjusted stock prices in weekly intervals

2.3 Rockwool's Market

Rockwool operates primarily in two segments: insulation and systems, with insulation accounting for 77% of their revenue and systems contributing 23% as seen in Figure 2. The company is renowned for producing stone wool, a major insulation material that represents their flagship product. According to Strait Research, the global insulation market is valued at 60 million USD and is expected to rise to over 100 million USD by 2030.⁶ Within this, the stone wool market is currently worth 16 million USD, projected to grow to over 25 million USD by 2030.⁷

Rockwool's main markets include Western Europe, which generates 59% of its revenue, Eastern Europe at 19%, and the rest of the world at 22% as shown in Figure 2. The company is strategically shifting its focus, reducing its presence in Russia and expanding in North America, aiming to increase its market share in the US and Canada. This geographical and market focus is elaborated in our strategic analysis report.⁸

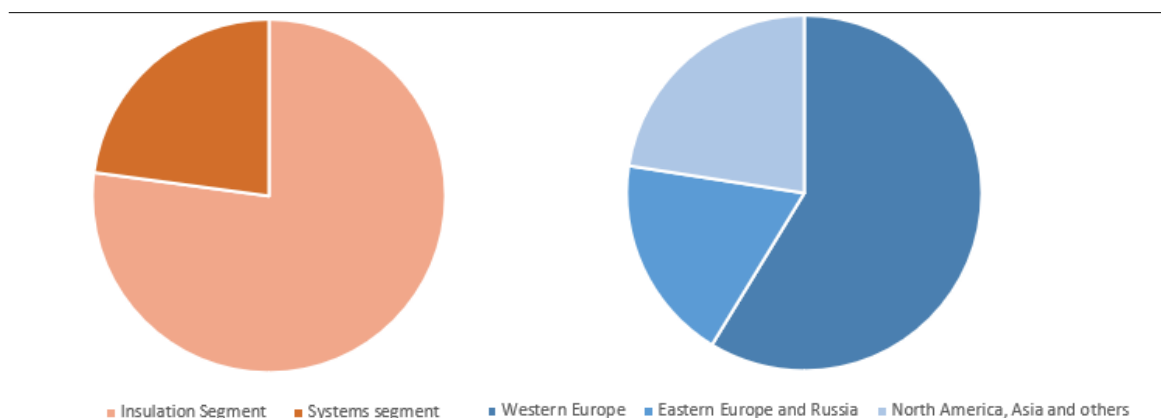
⁵Sustainability report 2023[8]

⁶Strait Research (2023)[5]

⁷Strait Research (2023)[4]

⁸Annual report 2023[7]

Figure 2. Rockwool sales division 2023



Source: Annual Report 2023⁹

3 Strategic Analysis

3.1 PESTEL

For a better understanding of which pros and cons might affect the business management of Rockwool A/S, we have to consider external factors. In the search of creating a comprehensive valuation and analysis of the company, a PESTEL-analysis is a crucial tool. The PESTEL-analysis consists of 5 key groups of external factor that can help us understand the economic climate of Rockwool; Political, Economic, Sociocultural, Technological and Environmental and Legal. For a full breakdown of the PESTEL-analysis refer to subsection A.1.

3.1.1 Political

The Russian invasion of Ukraine and subsequent Western sanctions pose significant challenges to Rockwool's operations, especially given its substantial market presence and workforce in Russia. Despite these geopolitical tensions, Rockwool's sales in Western markets have not yet declined, and its overall revenue in Eastern Europe and Russia has remained stable. However, the ongoing conflict introduces uncertainties and risks, including potential reputational damage and asset seizures. Rockwool's primary focus on the B2B sector in more politically stable regions mitigates some of these risks, allowing the company to maintain a level of operational stability amidst the turmoil.¹⁰¹¹¹²

3.1.2 Economic

The current global economic climate, influenced by factors like inflation and GDP growth, presents uncertainties that could potentially affect demand for real estate, impacting Rockwool's operations. However, according to the latest OECD forecasts projected in Table 1, the global economic outlook appears stable for the upcoming years. This stability suggests that these macroeconomic factors are not expected to pose significant risks to Rockwool's financial projections in the near future.¹³

¹⁰Rockwool statement on Russia[3]

¹¹Politico (2023)[15]

¹²UKRINFORM (2023)[9]

¹³OECD economic outlook (2024)[12]

Table 1. OECD's Real GDP and Inflation forecast of the G20-countries (2023-2033)

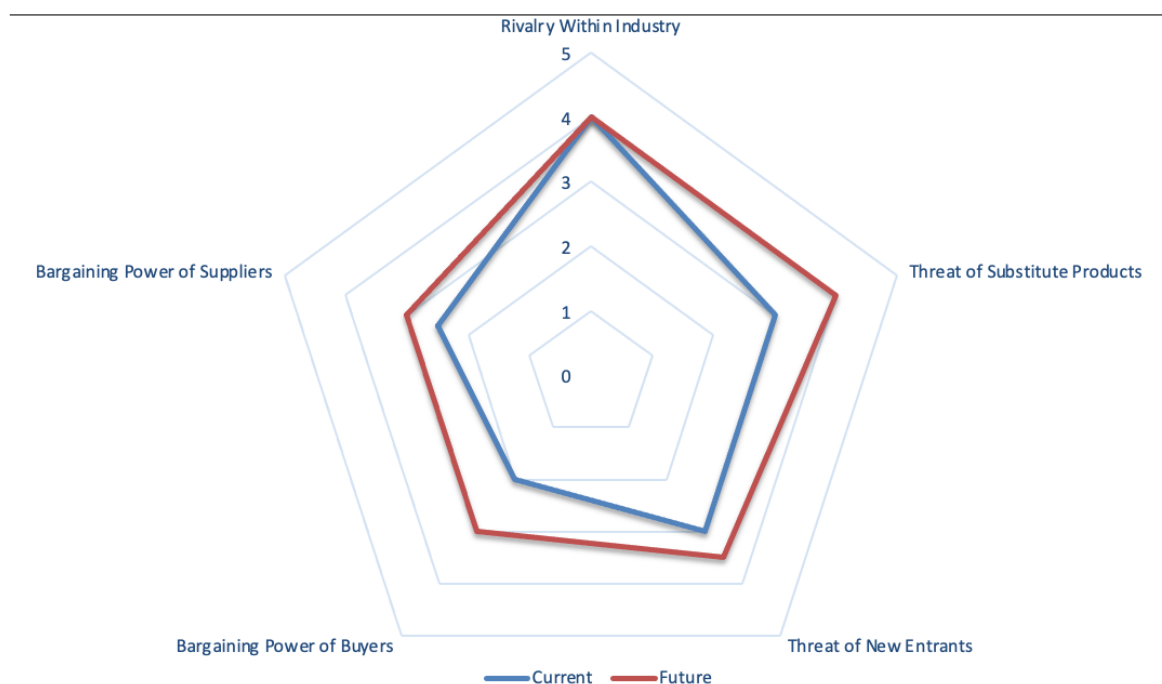
	2023	2024	2025	2026	2027	2028	2029	2030	2031	2032	2033
Real GDP Growth	3,16%	2,91%	3,08%	3,20%	3,09%	2,94%	2,81%	2,72%	2,66%	2,62%	2,58%
Global Inflation	5,03%	3,43%	2,00%	2,00%	2,00%	2,00%	2,00%	2,00%	2,00%	2,00%	2,00%

Source: OECD¹⁴

For the sociocultural, technological and the environmental and legal analysis we refer to subsection A.1.

3.2 Porter's Five Forces

To gain an insight into the competitive landscape of Rockwool A/S this section utilizes Porter's Five Forces framework (Porter, 2008)¹⁵. By analyzing forces such as industry rivalry, the threat of new entrants, the bargaining power of suppliers and buyers, and the threat of substitutes, this strategic analysis aims to uncover insights into Rockwool's market positioning and potential strategic moves. Rockwool's Porter's Five Forces analysis is summarized in Figure 3 and an in-depth analysis for each category can be found in subsection A.2.

Figure 3. A radar chart of Porter's Five Forces for Rockwool A/S.

Note: This radar chart illustrates the current as well as the future strength of each of Porter's Five Forces impacting Rockwool A/S, based on a scale from 1 (low influence) to 5 (high influence).

Source: Own calculations.

4 Financial Statement Analysis

This chapter revolves around the reformulation of the financial statements done according to the methodology of Penman (2013)¹⁶. The reformulation of the financial statements of Rockwool (2018-

¹⁵Porter (2008) [36]

¹⁶Penman (2013)[35]

2023) can be found in subsection A.3. The reformulation are used to do a profitability analysis of the historic financial performance of Rockwool.

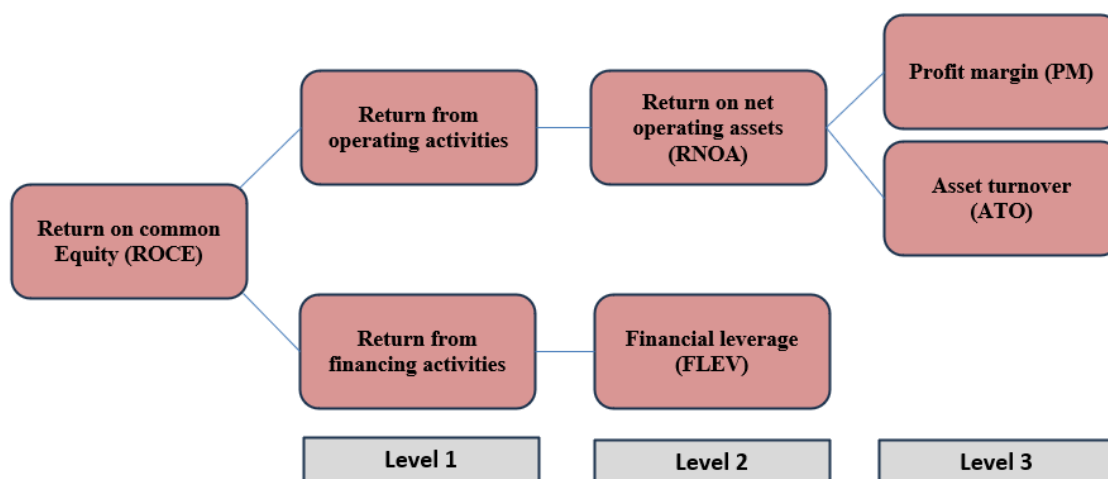
4.1 Profitability

In Stephen Penman’s 2013 work on financial statement analysis and security valuation, he emphasizes the importance of return on capital employed (ROCE) as a key metric in assessing profitability. ROCE measures how efficiently a company generates profits from the capital it uses in its operations. Specifically, it compares a firm’s operating profit to the capital invested in the business.

A higher ROCE indicates a more efficient use of capital, reflecting a firm’s ability to generate returns above the cost of that capital, which is crucial for long-term profitability and value creation. Penman explains that this ratio helps investors and analysts assess management’s effectiveness in deploying capital and whether the company is creating or destroying value. We use this information in the upcoming forecasting section by assessing the 3 levels of decomposition shown in Figure 4.

Specific calculation methods can be found in subsection A.5.

Figure 4. ROCE decomposition



4.1.1 Decomposition level 1

In this section, we analyze the level 1 ROCE decomposition for Rockwool, noting significant fluctuations in its performance. ROCE plummeted to 6.8% in 2020 from 26.1% in 2018, primarily due to the COVID-19 pandemic shown in Table 2. A recovery followed in 2021 as societies reopened, though levels have not returned to pre-pandemic figures, hindered further by the Russian invasion of Ukraine.

Further examination reveals Rockwool’s RNOA closely tracks ROCE, often exceeding it, reflecting a conservative strategy that avoids high debt levels and maintains substantial cash reserves. This resulted in a consistently negative FLEV, indicating a robust operational stance with significant financial stability. These indicators are generally perceived as positive but it also shows a potential lack in risk-taking and the seizing of business expanding projects that might prove costly in the future.

We also note that the huge fluctuations in SPREAD and NBC are contributed to the level of net financial liabilities close to zero - making these key figures very sensible.

Table 2. Decomposition level 1

Level 1	2018	2019	2020	2021	2022	2023
ROCE	26,1%	16,5%	6,8%	18,1%	11,5%	12,1%
RNOA	32,2%	19,0%	7,7%	18,8%	12,9%	12,4%
FLEV	-0,116	-0,090	-0,038	-0,018	0,001	-0,039
SPREAD	35,1%	20,6%	18,3%	33,6%	3444,9%	8,6%
NBC	-2,9%	-1,6%	-10,6%	-14,9%	-3432,0%	3,8%

Source: Annual Report (2018-2023) and own calculations

4.1.2 Decomposition level 2

Further analysis in Table 3 corroborates earlier findings: Rockwool's asset turnover and profit margin were strong pre-pandemic, suffered during 2020, and partially rebounded in 2021. Despite not fully recovering due to ongoing global economic challenges, the company maintains a healthy profit margin and asset turnover. Rockwool has consistently outperformed the construction market average, with a ratio close to 1 according to Vergo, indicating robust profitability despite differing operational structures in the sector.¹⁷

Table 3. Decomposition level 2

Level 2	2018	2019	2020	2021	2022	2023
RNOA	32,2%	19,0%	7,7%	18,8%	12,9%	12,4%
PM	9,3%	12,0%	5,9%	13,4%	8,2%	8,9%
ATO	3,46	1,58	1,30	1,41	1,58	1,39

Source: Annual Report (2018-2023) and own calculations

4.1.3 Decomposition level 3

The final section breaks down Rockwool's profit margin stability since 2018, highlighting minimal fluctuations in costs relative to profits. This stability underscores Rockwool's conservative operational approach and risk-averse business strategy. The core profit margin has remained steady, primarily influenced by external factors and broader economic conditions, reinforcing the company's focus on operational safety and limited strategic evolution. Overall, Rockwool's engagement in slow-moving, traditional markets, coupled with its conservative strategies, has ensured sustained, low-risk profitability. All of this will be considered in the upcoming forecasting section when deciding on a plausible growth rate.

¹⁷Vergo 2024[13]

Table 4. Decomposition level 3

Level 3	2018	2019	2020	2021	2022	2023
PM	9,31%	12,01%	5,91%	13,36%	8,17%	8,88%
Core PM	10,06%	10,42%	10,01%	9,99%	7,84%	10,62%
Gross profit	66,11%	66,50%	67,60%	63,93%	57,11%	64,18%
Delivery cost and indirect co	-14,35%	-13,74%	-13,92%	-14,16%	-13,21%	-12,02%
Other external cost	-8,65%	-7,86%	-7,06%	-6,63%	-6,96%	-8,25%
Personnel cost	-24,20%	-25,14%	-26,61%	-23,69%	-20,68%	-22,48%
D&A	-6,19%	-6,35%	-7,06%	-6,50%	-6,02%	-7,18%
Operating tax	-2,66%	-3,00%	-2,95%	-2,97%	-2,41%	-3,63%
Other Items PM	-0,75%	1,59%	-4,10%	3,36%	0,33%	-1,73%

Source: Annual Report (2018-2023) and own calculations

5 Forecasting

The next session includes a forecast estimation of the performance of Rockwool using the methodology of Penman (2013).¹⁸

In setting the duration of a forecast for financial valuations, defining the "extraordinary periods" is pivotal, though such periods are not explicitly detailed in Penman's methodology. Penman maintains a conservative stance, arguing that shorter forecasting periods are typically more reliable as they rely less on speculative projections and more on historical financial data. However, he acknowledges the flexibility necessary in forecasting, suggesting adjustments based on the specific financial conditions and strategic positioning of the firm in question.

For a company like Rockwool, which is entrenched in primarily the construction and manufacturing sector — a sector noted for its extended investment cycles and gradual market evolution — the length of the forecasting period is crucial for producing accurate and dependable valuation outcomes. Aswath Damodaran in his work supports this notion, advocating for a longer forecast horizon.¹⁹ Specifically, a 10-year period is deemed reliable for Rockwool to thoroughly account for the industry's dynamics and the company's strategic initiatives, which may influence financial outcomes over a more extended period. This approach aligns with the principle that the chosen forecast period should capture significant business developments, from growth through stabilization, which is especially relevant in industries characterized by long-term operations and investment cycles.

As for the forecasting itself, we use the latest consensus and analyst's report from Rockwool published in february 2024 as the main determinant of the future cash flows.²⁰ The projections are constructed by 20 analysts from various financial institution which makes for a credible estimates - at least for the period up until 2025. For the remaining years, we use a further analysis of the industry which Rockwool works within and attempt to estimate a robust forecast in consensus with the analysts estimates and general economical conditions.

For the period, we assume an overall stable economic environment.

5.1 Industry

Rockwool primarily generates revenue from specific sectors within the construction market, notably mineral wool manufacturing and insulation. In our forecasting analysis, we correlate their operational

¹⁸Penman (2013)[35]

¹⁹Damodaran (2006)[19]

²⁰Rockwool website (2024)[10]

involvement in the systems sector with a broader market defined as energy-efficient building market as information and analysis about the systems sector is limited.

Analysts predict a CAGR of 6.98% for the energy-efficient building industry from 2023 to 2028²¹ and a similar 7.2% CAGR for the insulation industry from 2023 to 2032²². However, the expected CAGR for mineral wool specifically is lower at 4.3% from 2023 to 2033²³. With 77% of Rockwool's revenue deriving from insulation and the remaining 23% mainly from efficient building products, we apply weighted averages of these growth rates to Rockwool's revenue segments. The broader insulation sector and the mineral wool sector is assumed evenly weighted within their insulation revenue stream. This results in an industry CAGR of 5.46% for Rockwool over the next 7-10 years, reflecting a blend of these market dynamics.

5.2 Revenue

When considering the forecasting of revenue for Rockwool in the upcoming 10 years, we use the numbers from the consensus and analysts' report for the prediction of the fiscal years 2024 and 2025 and largely rely on the industry CAGR found in the previous section for the period 2026-2030. In line with Penman's (2013) methodology²⁴, the growth will converge to the terminal growth rate of 3,5%. The reasoning behind this level of terminal growth can we found in subsection 5.4. Furthermore, we are aware that the growth level in 2023 lies significantly lower than our forecasted levels but deem the higher growth rates as realistic under the assumption of stabilizing macro economic tendencies. Our forecast for the revenue of Rockwool can be found in Table 5.

Table 5. Revenue forecast of Rockwool A/S (2023-2033)

	Avg. 2018-2023	2023	2024	2025	2026	2027	2028	2029	2030	2031	2032	2033
Revenue		3.635	3.600	3.778	3.984	4.202	4.431	4.673	4.928	5.165	5.379	5.568
CAGR	6,27%	-7,32%	-0,10%	4,90%	5,46%	5,46%	5,46%	5,46%	5,46%	4,81%	4,14%	3,50%

Sources: Rockwool consensus & analysts' report 2023 [10] and own calculation.

5.3 Profit Margin and Asset Turnover

Our forecast for Rockwool's core profit margin is straightforward. From 2019 to 2023, the margin remained steady between 9.99% and 10.62%, except for an outlier year in 2022. Based on the consensus and analysts' report for 2024/25, we project a core profit margin of 10.5% in 2024 and 10.77% in 2025, assuming it will align with 2023 levels over the next decade.

Similarly, we expect Asset Turnover (ATO) to remain stable, having fluctuated between 1.30 and 1.58 since 2019, with a 2023 level of 1.39. We deem this as a realistic level for the future. While we could speculate on slight declines in ATO and profit margins due to a reduced CAGR as the period approaches the terminal growth rate, we lack concrete evidence to justify deviating from historical consistency. Thus, we maintain a forecast of stable financial metrics based on data since 2019 - in particular the latest year of 2023.

²¹Globe Newswire (2024)[38]

²²Precedence Research (2023)[39]

²³Futute Market Insights (2023)[41]

²⁴Penman (2013)[35]

Table 6. Core Profit Margin and ATO Forecast of Rockwool A/S (2023-2033)

	2023	2024	2025	2026	2027	2028	2029	2030	2031	2032	2033
Core PM	10,62%	10,50%	10,77%	10,62%	10,62%	10,62%	10,62%	10,62%	10,62%	10,62%	10,62%
ATO	1,39	1,39	1,39	1,39	1,39	1,39	1,39	1,39	1,39	1,39	1,39

Sources: Rockwool consensus & analysts' report 2023 [10] and own calculation.

5.4 Terminal Growth Rate

In the valuation process, assuming a constant terminal growth rate is a common method, reflecting the concept that a company will continue operating indefinitely. According to the approach discussed by Penman in 2013²⁵, it's generally accepted that no company can sustain a growth rate that exceeds the nominal growth rate of the economy where it operates over the long term. Hence, it's typical to set the terminal growth rate at or below the expected long-term nominal GDP growth rate to avoid unrealistic valuations.

For a company like Rockwool, which operates on a global scale, it is appropriate to consider the global economic outlook. The OECD's 2024 forecast indicated a decrease in trend real GDP growth from 3% to 1.5% by 2060, mainly due to a slowdown in major emerging markets²⁶. Combining this with the Federal Open Market Committee's and the ECB's inflation target of 2%²⁷, a nominal terminal growth rate of 3.5% is used in our financial analysis to align with these projections. This approach ensures that the company's growth assumptions are grounded in realistic expectations of global economic conditions.

6 Cost of capital

For Rockwool A/S, understanding the cost of capital is essential to evaluate the valuation. The Weighted Average Cost of Capital (WACC) represents the average rate of return that a company is expected to pay to its security holders to finance its assets. The WACC is calculated using the formula:

$$WACC = \frac{E}{V}r_e + \frac{D}{V}r_d(1 - \tau_C)$$

where E represents the market value of the equity, D the market value of the debt, V the total value ($E + D$), r_e the cost of equity, r_d the cost of debt, and τ_C the corporate tax rate.

According to Damodaran (2012)²⁸, the precision of WACC calculations significantly influences valuation accuracy, mentioning the importance of correctly estimating each component. Koller et al. (2010)²⁹ argue for careful testing of the underlying assumptions used in the cost of equity and debt calculations to ensure they reflect market realities and firm-specific risks. They highlight the impact of market conditions and financial strategy on WACC, suggesting that adjustments may be necessary to address these dynamic factors.

The subsections that follow will discuss and evaluate these methodologies for estimating the cost of debt, including the risk-free rate and company-specific premium, and the cost of equity through market risk premium and beta estimation, as recommended by Damodaran (2012), and Koller et al. (2010).

²⁵Penman (2013)[35]

²⁶OECD economic outlook (2024)[12]

²⁷European Central Bank website[14]

²⁸Damodaran (2012) [18]

²⁹Koller et al. (2010) [32]

6.1 Cost of Debt

The cost of debt (r_d) reflects the effective rate that the company must pay on its current debt to finance its assets. The after-tax cost of debt is particularly necessary from a valuation perspective because interest expenses are tax-deductible, reducing the net cost to the company.

The formula to calculate the after-tax cost of debt is given by:

$$r_d = (r_f + r_s)(1 - \tau_C) \quad (1)$$

where r_d is the cost of debt, τ_C represents the corporate tax rate, r_f is the risk-free rate, and r_s is the company specific premium. This equation underscores the influence of both the market conditions, encapsulated by r_f and r_s , and the tax environment, indicated by τ_C , on the firm's debt cost.

In the following subsection, we will explore how Rockwool A/S manages its financing costs in the context of prevailing interest rates and its risk profile, influenced by company-specific factors and market perceptions of credit risk. According to Berk and DeMarzo (2016)³⁰, an accurate assessment of the risk-free rate and risk premium is essential for determining the true cost of debt, as these factors directly affect corporate borrowing costs.

6.1.1 Risk-free Rate

The selection of an appropriate risk-free rate is a central component in the calculation of the cost of debt and equity in financial modeling. According to academic theories, a risk-free rate is theoretically the return on investment with zero risk, typically represented by government securities of a stable financial system according to Koller et al. (2010). For an investment to qualify as risk-free, it must meet two key criteria: the absence of default risk and no reinvestment risk, which can be approximated by using the yield of a zero-coupon government bond matching the investment's duration [16]. In practice, the choice of a risk-free rate often defaults to government bonds of the country where the analysis is conducted, adjusted to match the currency and duration of the cash flows being discounted. Holm et al. (2005)³¹ and Sørensen (2021)³² support using local government debt instruments as they reflect local economic conditions and currency risk. Given this context, Danish government bonds, particularly the 10-year bond, are considered a solid benchmark for the risk-free rate in Denmark due to the country's economic stability and low default risk.

³⁰Berk and DeMarzo (2016) [16]

³¹Holm et al. (2005) [29]

³²Sørensen (2021) [42]

Figure 5. Historical annual return for the 10-year Danish Government Bond.



Note: Calculated averages for 10y, 15y, and 20y based on own calculation, monthly data.
Source: DST.dk & MPK3 [40]

Table 7. Calculated averages for the 10-year Danish government bond.

	Spot	10-year average	15-year average	20-year average
Rate	2.03%	0.57%	1.17%	1.88%

Note: The rates are calculated based on historical data.

Table 7 summarizes the historical yields for the 10-year Danish government bond shown in Figure 5. The current 10-year spot rate of 2.03% provides a contemporary measure of the risk-free rate, capturing the present economic conditions. Academic perspectives such as EY (2022)³³ and Fernandez (2020)³⁴ validate the use of the spot rate of long-term government bonds as a real-time, empirical risk-free rate for financial evaluations. This approach is consistent with the principle of matching the duration of the risk-free rate to that of the forecast period in valuation models.

Hence, employing the 10-year Danish government bond spot rate as the risk-free rate for evaluating Rockwool A/S's cost of capital is aligned with both academic and practical financial standards.

6.1.2 Company Specific Premium

The Company Specific Premium (CSP) refers to the additional risk premium that investors demand to compensate for risks unique to a particular company, which are not captured by the market-wide risk factors. This premium is often reflected in the cost of debt as lenders demand higher returns to offset higher risks associated with specific firms.

³³EY(2022) [25]

³⁴Fernandez (2020) [27]

For Rockwool A/S, the calculation of the Interest Coverage Ratio (ICR) is an indicator used to assess the company's ability to meet its debt obligations. The ICR is calculated as follows:

$$ICR = \frac{EBIT + \text{Leasing Costs}}{\text{Finance Costs} + \text{Leasing Costs}} = \frac{518 + 87}{34 + 87} = 5.00$$

An ICR of 5.00 for a large-capitalization company corresponds to a relatively high credit rating, of A2/A rating, which indicates a strong ability to meet financial commitments. A credit rating of A2/A corresponds to a default spread of 1.07% according to Damodaran (2024J)³⁵.

Table 8. Public credit ratings of competitors.

Firm	Moody's	S&P
Saint-Gobain	Baa1	BBB
Owens Corning	Baa1	BBB

Table 8 assessment helps in evaluating the cost of debt for Rockwool A/S, as it provides a basis for comparison against the public credit ratings of its competitors. Unfortunately, we could only find ratings for two competitors. We observe that Rockwool A/S has a significantly higher credit rating than its competitors which aligns with its debt-to-equity ratio touched upon in section 4.

By inserting the derived components found above into Equation 1 we can now calculate the cost of debt r_d .

$$r_d = (r_f + r_s)(1 - \tau_C) = (2.03\% + 1.07\%) \cdot (1 - 22.00\%) = 2.42\%$$

Therefore this paper proceeds with an after-tax cost of debt estimated to be 2.42%.

6.2 Cost of Equity

6.2.1 Capital Asset Pricing Model

Determining the cost of equity is crucial for assessing investment viability and corporate valuations. In financial theory, the Capital Asset Pricing Model (CAPM) serves as a fundamental tool for estimating the cost of equity. This model asserts a linear relationship between the expected return on a security and its systematic risk. The CAPM formula, central to this approach, is defined as:

$$r_e = r_f + \beta(E(r_m) - r_f) \quad (2)$$

where r_e is the cost of equity, r_f is the risk-free rate, β measures, in this case, Rockwool's stock price volatility relative to the market, and $E(r_m) - r_f$ represents the expected market risk premium. A beta above one suggests higher volatility compared to the broader market. Companies with a high risk level usually obtain a high beta value making the companies vulnerable to fluctuations in the market. The market risk premium ($E(r_m) - r_f$) is the extra return, deducted the risk-free rate, that investors require to invest in the market instead of risk-free assets, thus the market risk premium.

The application of CAPM assumes market efficiency and rational behaviors among investors. These assumptions often come under critique in the real-world application of the model. Fama and French (2004)³⁶ highlights that CAPM does not account for empirical irregularities observed in market behaviors, which could be influenced by factors beyond market risk. Meanwhile, Pratt and

³⁵Damodaran (2024J) [20]

³⁶Fama and French (2004)[26]

Grabowski (2011)³⁷ suggest that while CAPM is useful, adjustments might be necessary to capture additional dimensions of risk in practical scenarios. Furthermore, Berk and DeMarzo (2016)³⁸ provide a comprehensive exploration of CAPM, reinforcing its foundational role in modern financial theory but also acknowledging its limitations due to its assumption-based framework.

Despite these criticisms, the simplicity of CAPM makes it a key indicator in financial analysis, providing a starting point for understanding risk-return trade-offs in equity investments. It is instrumental in the calculation of the WACC, integrating the cost of equity to provide a sharp view of a company's capital costs but it is not the only existing model.

6.2.2 Arbitrage Pricing Theory Model

The Arbitrage Pricing Theory (APT) model, developed as an alternative to the CAPM, incorporates multiple risk factors to address some of the limitations found in CAPM. APT suggests that the expected return of a financial asset can be modeled as a linear function of various macroeconomic factors or theoretical market indices, each associated with its own specific risk premium. This differs from CAPM, which relies on a single market risk factor (β), by allowing for the inclusion of diverse factors such as inflation rates, GDP growth rates, interest rates, and other variables that could impact asset prices. While the APT provides a more flexible approach by considering multiple risk factors, CAPM remains widely favored for several reasons. Firstly, CAPM's single-factor model offers a straightforward calculation method that is far easier to implement than the potentially complex multi-factor APT model. Despite the known limitations highlighted by Fama and French (2004) and others, CAPM is chosen for our paper. Its balance of simplicity and widespread recognition makes it particularly suitable for our analytical needs, offering a robust framework for evaluating Rockwool A/S's cost of equity.

6.2.3 Market Risk Premium

In the subsequent sections, we will explore three primary methods for estimating the market risk premium as described in Damodaran (2024)³⁹. Each method offers distinct insights and methodologies for understanding and quantifying this crucial element of the CAPM.

6.2.4 Survey Market Risk Premium

The Survey Market Risk Premium (MRP) approach involves using risk premiums that investment professionals forecast based on their expectations and experiences. According to the recent academic work by Fernandez et al. (2024)⁴⁰ the expected current market risk premium for Denmark is 5.8%, and the risk-free rate is 2.9%. Adjusting this method to utilize the spot 10-year Danish government bond rate of 2.03% as the risk-free rate, we calculate the Survey MRP as follows:

$$\text{Survey MRP} = E(r_m) - r_f = 5.8\% - 2.03\% = 3.77\%$$

This calculation adjusts the general expected MRP by incorporating specific, current risk-free rate data, providing a localized view of the premium. While the survey method of determining the market risk premium provides real-time insights into market expectations, it comes with notable limitations. As discussed by Damodaran (2024)⁴¹, this approach can introduce subjective biases, as

³⁷Pratt and Grabowski (2011)[37]

³⁸Berk and DeMarzo (2016)[16]

³⁹Damodaran (2024)[22]

⁴⁰Fernandez et al. (2024)[28]

⁴¹Damodaran (2024)[22]

it relies heavily on the personal expectations and experiences of finance professionals, which may not accurately reflect broader market realities. Furthermore, the survey results can vary significantly depending on the economic outlook and sentiment at the time of the survey.

Due to these potential inaccuracies and the subjective nature of the data collected, we will also investigate the implied market risk premium approach.

6.2.5 Implied Market Risk Premium

The concept of implied market risk premium can be effectively demonstrated using the Dividend Discount Model (DDM), a fundamental valuation approach in financial analysis. According to Damodaran (2024)⁴², the value of equity is calculated as the present value of expected future dividends. This model is particularly useful when dividends are projected to grow at a constant rate indefinitely, leading to the formulation of the stable growth DDM:

$$\text{Value of Equity} = \frac{\text{Expected Dividends Next Period}}{\text{Required Return on Equity} - \text{Expected Growth Rate}}$$

This formula simplifies the valuation of equity to the present value of an infinite series of dividends growing at a stable rate. Following the basic framework of the Dividend Discount Model Damodaran (2024)⁴³ expands the model to:

$$\text{Value of Equity} = \sum_{t=1}^N \frac{E(FCFE_t)}{(1 + k_e)^t} + \frac{E(FCFE_{N+1})}{(k_e - g_N)(1 + k_e)^N} \quad (3)$$

where $E(FCFE_t)$ represents the expected free cash flow to equity in year t , k_e is the required return on equity, g_N is the growth rate assumed to perpetuate after year N . N is the number of periods over which the cash flows are explicitly forecasted.

This method is grounded in the assumption that dividends will grow perpetually at a predictable rate, which encapsulates investor expectations about the firm's long-term profitability and growth potential. Such a model is crucial in deriving the implied market risk premium, which reflects the extra return over the risk-free rate that investors require to compensate for the risk undertaken by investing in the market.

Continuing with the methodology outlined by Damodaran (2024)⁴⁴, we apply his approach to calculate the implied market risk premium as of January 1, 2024, using data from the S&P 500, due to the insufficient data on cash payouts for the MSCI World Index. The earnings and cashflows underneath are set to grow at 3.38% per year forever equal to the risk free rate. The market risk-free rate, calculated based on the S&P500, is projected to represent the market risk-free rate for the broader market making it relevant for our paper. Ideally we would have found an implied MRP using the same risk-free rate as for the cost of debt but insufficient data on the C25-index limits us to the use of the S&P500-index.

During 2022, financial markets adjusted to higher inflation levels, leading the treasury bond rate to climb to 3.88%, marking the end of a period characterized by exceptionally low-interest rates. This surge in rates and ongoing economic uncertainties initially drove stock prices down by about 18%. However, the market showed remarkable resilience in 2023, bouncing back amid persistent concerns about inflation and economic growth. Given the S&P500 index value of 4,769.83 as of January 1, 2024, and the assumption that earnings growth after the terminal year will continue at a rate of

⁴²Damodaran (2024)[22]

⁴³Damodaran (2024)[22]

⁴⁴Damodaran (2024)[22]

Table 9. Yearly Earnings, Cash Payout Percentage, and Cash Returns (Data for S&P 500)

	Last 12 months	2024	2025	2026	2027	2028	Terminal Year
Earnings	219.70	238.89	259.76	282.45	307.13	333.96	346.92
Cash Payout (%)	77.85%	77.85%	77.85%	77.85%	77.85%	77.85%	77.85%
Cash Return	164.25	185.97	202.21	219.88	239.09	259.97	270.06

Note: All values are in USD. Cash return is defined as dividends plus buybacks.

Source: Damodaran (2024)⁴⁵. See subsection A.4 for definitions.

3.88% which matches the risk-free rate based on the 10-year US government bond, we formulate the following equation to calculate the expected return on equity:

$$4,769.83 = \frac{185.97}{(1+r)} + \frac{202.21}{(1+r)^2} + \frac{219.88}{(1+r)^3} + \frac{239.09}{(1+r)^4} + \frac{259.97}{(1+r)^5} + \frac{270.06}{(r - 0.0388)(1+r)^5}$$

Upon solving this equation, the required return on equity, r , is found to be 8.48%. By deducting the risk-free rate, notably the treasury bond rate of 3.88%, we calculate the implied market risk premium to be

$$\text{Implied MRP} = 8.48\% - 3.88\% = 4.60\%$$

By January 1, 2024, these dynamics translated into an implied market risk premium (MRP) of 4.60%, as illustrated in the calculations of Damodaran (2024). It's important to note that the treasury bond rate held steady at 3.88% from the previous year, and an increase in stock prices reduced the expected return on stocks to 8.48%. Subtracting the treasury bond rate from this figure gives us the implied MRP of 4.60%, a notable decrease from 5.94% earlier in 2023⁴⁶.

With Rockwool A/S international market shares⁴⁷, we find it reliable to use the MRP calculated from S&P500.

6.2.6 Historical Market Risk Premium

The historical premium approach involves comparing the actual returns earned on stocks over a long period with the return on default-free security, here the 10-year US government bond. The difference between these returns, calculated annually, represents the historical risk premium. In this section, we will use broad indexes, namely the MSCI World Index and S&P500 to assess the market risk premium following the same approach as of Damodaran, 2024⁴⁸.

We find the historic market risk premium to be 5.35% using the MSCI World index of 50 years. As this premium is not used in the final valuation, we refer to subsection A.6 for a more in-depth analysis and discussion.

6.3 Estimation of Beta

Estimating the beta (β), a measure of stock volatility compared to the market, is crucial for Rockwool A/S as it directly influences the company's WACC. Beta helps determine the risk premium used in the CAPM, which in turn calculates the required return on equity.

For Rockwool, which operates in the economically sensitive building materials industry, an accurate beta reflects how external economic shifts might impact the company. This estimation involves

⁴⁶Damodaran (2024)[22]

⁴⁷Market Screener[33]

⁴⁸Damodaran (2024)[22]

analyzing Rockwool’s historical stock performance against a broad market index, here using the MSCI World Index and S&P500, two relevant market index that captures industry-specific risks and broader market conditions.

6.3.1 Beta estimation with top-down approach

The top-down approach is widely used and expressed through the CAPM which is a foundational tool in financial analysis used to determine the expected return on an investment based on its risk relative to the market. The CAPM equation is for the beta analysis expressed as:

$$r_i = \alpha + \beta \times r_m + \epsilon,$$

where r_i represents the returns of Rockwool A/S, which are the focus of this analysis; α acts as the y-intercept in the regression analysis, depicting returns independent of the market movements; β measures Rockwool’s volatility compared to the market; r_m denotes the returns of the market index and ϵ captures all other factors affecting Rockwool’s returns that are not explained by market movements, that is the error term. For a robust and reliable beta estimation, Koller et al. (2010) ⁴⁹ argues it is customary to utilize monthly returns, with a 5-year period as a minimum. This duration ensures a substantial number of data points, enhancing the reliability of the beta calculation and minimizing potential biases that shorter intervals might introduce. Utilizing monthly rather than daily or weekly data also helps reduce the noise and systematic biases that can distort the beta estimation.

By applying this approach, we can establish a consistent and methodologically sound basis for evaluating Rockwool’s systematic risk as it relates to broader market dynamics. By regressing Rockwool’s monthly returns against those of the MSCI World Index over the past five years, we calculate the beta (β) as follows:

$$\beta = \frac{cov(r_i, r_m)}{var(r_m)} = 1.44$$

this β indicates a notable level of volatility relative to the market. This higher beta value shows that Rockwool’s stock is more sensitive to market movements than an average equity, exhibiting a risk profile that exceeds the market’s average volatility. To strengthen the assumption of S&P500 being used as an index for the broader market we’ve repeated the calculation above. The results are shown in Table 10.

Table 10. Rockwool A/S’s beta values

Index	Beta-value
MSCI World Index	1.44
S&P 500	1.45

Note: The values are calculated as a regression shown in the equation above based on monthly returns from MSCI and S&P 500 for a 5-year period.

As seen in the table the calculated beta is almost identical strengthening the validation of S&P 500 as a broader market index. Nonetheless, we will continue with the beta based on the MCSI World Index.

⁴⁹Koller et al. (2010) [32]

To visually capture the dynamics of Rockwool's beta over time, a graph of the rolling beta estimates has been included. This graph provides a visual illustration of how Rockwool's beta fluctuates over sequential time windows, highlighting periods of increased or decreased market sensitivity.

Figure 6. Rolling Beta Estimates



Note: Valuation as of December 31, 2023.

Source: Calculated 1y, 2y, 3y and 4y based on own calculation.

The graph illustrates a relatively stable beta estimate exceeding 1 for Rockwool A/S over the observed period. However, it is noteworthy that within the one-year rolling estimate, the beta dipped below 1, and there was even an instance where the beta approached negative values and values below 1. This instance highlights potential periods of atypical market behavior or company-specific events affecting Rockwool's stock volatility differently from broader market movements.

6.3.2 Beta Estimation with Bottom-Up Approach

The beta estimation process using a bottom-up approach allows for a more nuanced analysis of Rockwool A/S's risk profile, as compared to the traditional top-down approach. This method aggregates the betas of peer companies to derive Rockwool's overall beta, offering a granular view that aligns closely with Rockwool's operational realities. According to Koller et al. (2010)⁵⁰, this approach is not only suited for private companies but is also advantageous for public firms, especially when individual company betas may be distorted by specific periods or non-systematic volatility.

$$\beta_U = \frac{\beta_L}{1 + (1 - \tau) \left(\frac{D}{E} \right)}$$

As Damodaran (2009)⁵¹ advocates, utilizing peer-estimated betas can result in more precise measures than those derived from single regression analyses due to the reduced standard error associated with bottom-up estimates. The approach assumes that companies in the same industry face comparable operational risks, hence the beta of Rockwool is calculated as a weighted average of its peers' betas, after adjusting for differences in capital structure.

⁵⁰Koller et al. (2010) [32]

⁵¹Damodaran (2009) [17]

Table 11. Peer D/E

	Peer Group	Market Value	D/E	Effective Tax Rate
Rockwool		42,138	4.10%	25.5%
Saint-Gobain	<i>Building Materials</i>	33,815	65.47%	27.7%
Owens Corning	<i>Building Materials</i>	14,189	62.82%	24.5%
Kingspan Group	<i>Building Materials</i>	14,394	54.16%	17.7%
CSR Limited	<i>Building Materials</i>	1,191	11.90%	25.4%
Patrick Industries, Inc	<i>Building Materials</i>	2,454	115.5%	23.20%
STO SE & CO. KGAA	<i>Building Materials</i>	898	14.21%	32.68%

Note: Data based on latest reports (EURm)

Sources: Yahoo Finance as of 12/31/2023

We selected peer companies within the building materials industry, reflecting similarities in market dynamics and financial risks. The peer group includes firm presented in Table 12, representing a cross-section of the industry with varying sizes and market scopes. The levered and unlevered betas of these companies are summarized in Table 12, calculated over a 5-year period using monthly returns against the MSCI World Index.

Table 12. Peers levered and unlevered beta values for Rockwool A/S's competitors

Company	Levered β_L	Unlevered β_U
Saint-Gobain	1.41	0.95
Owens Corning	1.47	1.00
Kingspan Group	0.98	0.68
CSR Limited	0.80	0.74
Patrick Industries, Inc	1.73	0.92
STO SE & CO. KGAA	0.99	0.90
Median	1.20	0.87
Average	1.23	0.91

Note: Levered β is based on company annual return against MSCI World Index annual return.

Source: Yahoo Finance and own calculations

The bottom-up approach to beta estimation underscores the systematic and financial risks embedded in Rockwool A/S's industry sector. By calculating the median and average of peer betas, we remove the influence of outliers and provide a robust estimate for Rockwool's systematic risk. According to Damodaran (2012)⁵², the median is often preferred over the average in financial metrics due to the discard of potential outliers. We do not observe any significant outliers, and we therefore proceed with the average beta estimate. The final levered beta, adjusted for Rockwool's specific tax rate and debt-to-equity ratio, reflects a comprehensive measure of risk-adjusted for industry-specific factors. Given the necessity to adjust for Rockwool A/S's specific financial structure, the levered beta is recalculated from the base of the average unlevered beta, which is 0.91, according to peer group analysis. The effective tax rate (τ) is 25.50% and the debt-to-equity ratio (D/E) is 4.10% for Rockwool, based on data from the 2023 Annual Report.

Applying the re-levering formula:

$$\beta_L = \beta_U \times \left(1 + (1 - \tau) \times \left(\frac{D}{E} \right) \right)$$

⁵²Damodaran (2012) [18]

we substitute the values:

$$\beta_L = 0.91 \cdot (1 + (1 - 25.5\%) \cdot (4.10\%)) = 0.94$$

Thus, the levered beta of Rockwool A/S, reflecting both the operational and financial risks, is calculated to be 0.94. This levered beta provides a measure of Rockwool’s total risk as perceived by the market, incorporating both its non-existent debt level and operational volatility and the average unlevered beta from its competitors. Rockwool A/S exhibits a notably low debt-to-equity ratio of 4.10% which is significantly lower than that of its industry peers. This minimal leverage, according to Damodaran (2012), reduces financial risk, which typically results in a lower beta. However, the observed beta values of 1.44 and 0.94 suggest that other factors might be contributing to a relatively high beta despite the low leverage.

Following a comparative analysis of Rockwool A/S and its competitors, and considering that Rockwool maintains a notably low debt-to-equity ratio of 4.10%, this paper will adopt the beta estimate of 1.44 derived through the top-down approach for further financial evaluations.

6.3.3 Comparison with Professional Estimates

In assessing the robustness of our beta estimation for Rockwool A/S we conducted a review of equivalent metrics calculated by leading financial institutions as of May 2024. This comparative analysis is crucial to validate our findings within the broader financial community’s consensus.

Table 13 aggregates beta estimates reported by four prominent financial services, with each institution utilizing potentially different methodologies, indices, and review periods, which might explain some discrepancies in their results. The beta values should, therefore, be interpreted with caution due to the lack of information regarding the periods and data sets used. However, despite these variations, there is a noteworthy alignment in the findings. The average beta calculated from these diverse sources is 1.45, which is in the range of our estimate by the regression approach, reinforcing the reliability of our analytical approach.

Table 13. Beta Values as Reported by Diverse Financial Institutions

Professional Institution	Beta
Yahoo Finance	1.29
Infront Analytics	1.20
Financial Times	1.65
Reuters	1.66
Average	1.45

Note: Estimates valid as of May 30, 2024

Sources: Yahoo Finance (2024), Infront Analytics (2024), Financial Times (2024), Reuters (2024).

Incorporating the beta found in subsection 6.3.1 into Equation 2, enables us to calculate Rockwool A/S’s cost of equity as follows:

$$r_e = 2.03\% + 1.44 \cdot 4.60\% = 8.65\%$$

7 Weighted Average Cost of Capital

The Weighted Average Cost of Capital (WACC) for Rockwool A/S is computed based on the cost of debt and equity derived in the previous sections.

Rockwool A/S is a publicly traded company, and therefore, the market value of equity is derived by multiplying the stock price by the number of outstanding shares. According to the annual report as of December 31, 2023, the closing stock price was DKK 1,976 with 21,39m outstanding shares, yielding an equity value of DKK 42,266m. The market value of debt, which reflects Rockwool's net financial obligations, is reported as DKK -1,592m (Rockwool, 2023). Using the cost of equity r_e of 8.65% as calculated and a cost of debt r_d of 2.42%. The WACC is computed as follows:

$$WACC = \frac{E}{V} \times r_e + \frac{D}{V} \times r_d \times (1 - \tau_C)$$
$$WACC = \frac{42,266}{42,266 - 1592} \cdot 8.65\% + \frac{-1592}{42,266 - 1592} \cdot 2.42\% = 8.90\%$$

Inserting these values into the WACC formula yields a result of 8.90%, which will be applied in the following valuation section. The tax component is already incorporated in the cost of debt.

8 Valuation

This section aims to determine the fundamental value of Rockwool A/S using the Discounted Cash Flow (DCF) model. As a widely acknowledged valuation tool in both academic and professional practice⁵³, the DCF model offers a great measure of a company's value by forecasting its future cash flows and discounting them to present value. In addition to the DCF analysis, this valuation will incorporate a sensitivity analysis to address the uncertainties associated with various valuation inputs. Furthermore, we will relate a multiple valuation to the DCF valuation.

8.1 Discounted Cash Flow Model (DCF)

The Discounted Cash Flow (DCF) approach is central to our valuation methodology for Rockwool A/S. This technique estimates the value of a company by computing the present value of projected future Free Cash Flows to the Firm ($FCFF = OI - \Delta NOA$), adding the terminal value that reflects the cash flows beyond the forecast period. The DCF formula employed in our analysis is expressed as follows:

$$DCF = \sum_{t=1}^n \frac{FCFF_t}{(1 + WACC)^t} + \frac{FCFF_{n+1}(1 + g)}{(WACC - g)} \times \frac{1}{(1 + r_{WACC})^n}$$

Here, $FCFF_t$ represents the free cash flow to the firm in year t , g denotes the perpetual growth rate, and n is the number of forecasted periods.

Utilizing financial data forecasted over a ten-year period, the enterprise value (EV) of Rockwool is calculated, along with the terminal value considering perpetually growing cash flows beyond the forecast horizon.

⁵³Koller et al. (2010) [32]

Figure 7. Discounted Cash Flow Valuation

	2023	2024	2025	2026	2027	2028	2029	2030	2031	2032	2033
Sales	3.635	3.600	3.778	3.984	4.202	4.431	4.673	4.928	5.165	5.379	5.568
OI after tax	323	320	336	354	373	394	415	438	459	478	495
NOA	2.593	2.580	2.706	2.854	3.010	3.174	3.348	3.531	3.700	3.854	3.988
RNOA	12,4%	12,4%	12,4%	12,4%	12,4%	12,4%	12,4%	12,4%	12,4%	12,4%	12,4%
FCF=OI-ΔNOA	333	209	206	217	229	242	255	289	289	325	360
Discount factor (WACC 8.68%)	1,09	1,19	1,29	1,41	1,53	1,67	1,82	1,98	2,15	2,35	
Present value (PV) of FCF		306	176	160	155	150	145	140	146	151	153
Sum of PV(FCF)	1.682										
WACC	8,90%										
Terminal value (TV) (TG 3.5%)	3,5%										6.894
Present value of TV	2.939										
Enterprise value	4.621										
NFO	-221										
Minority interest	-										
Equity value for Common Shareholders	4.842										
# of share in millions	21,37										
Price per share EUR	226,54										
EURO/DKK exchange rate 12/31/2023	7,5429										
Price per share DKK	1.709,13										

Note: Valuation as of December 31, 2023. EURm

Source: Rockwool A/S Annual Report (2019, 2020, 2021, 2022, 2023) and own calculations.

The calculated enterprise value for Rockwool, based on the DCF model, stands at EUR 4,621 million. After adjusting for net financial obligations (NFO) of EUR -221 million and accounting for minority interest, the equity value for common shareholders is determined to be EUR 4,842 million. With the current number of outstanding shares, the estimated price per share is approximately DKK 1,709.13 on December 31, 2023, thus below the listed stock price of DKK 1,976.

This valuation underscores the intrinsic value of Rockwool. The application of a sensitivity analysis further enhances the reliability of our valuation, accommodating potential variations in critical assumptions such as WACC and growth rates.

8.2 Sensitivity Analysis

Sensitivity analysis is a tool in financial modeling and valuation, providing insights into how changes in key assumptions impact the valuation outcomes. By systematically varying these assumptions, we can assess the robustness of our valuation and identify the factors that most significantly influence the results. Considering that 63.6% of the enterprise value is derived from cash flows determined by the terminal growth rate, it is imperative to thoroughly investigate the impact of changes in the parameters of the cost of capital.

In our valuation of Rockwool, we conducted a sensitivity analysis by varying the Market Risk Premium (MRP), WACC, and beta in Figure 8. This analysis allows us to observe how changes in these parameters affect the company's valuation. As illustrated in the figure, if we had used the peer-estimated beta value of 0.94, resulting in a WACC of 6.12%, we would have obtained a significantly higher share price of DKK 2,895.52. Utilizing the calculated historical risk-free premium of 5.35% would yield a share price of DKK 1,530.55. Furthermore, we observe that the beta value has a significant impact on the calculation of the WACC due to Rockwool's net financial obligation status. Both the beta and the risk-free rate significantly impact the cost of equity; therefore, we will further examine the sensitivity of the risk-free rate.

Figure 8. Sensitivity analysis with regards to MRP, beta, and WACC

		Market Risk Premium (MRP)								
		4,20%	4,30%	4,40%	4,50%	4,60%	4,70%	4,80%	4,90%	5,35%
6,12%	0,94	2.895,52	2.818,53	2.746,90	2.680,07	2.617,58	2.559,03	2.504,04	2.452,31	2.252,95
7,11%	1,14	2.328,39	2.275,71	2.226,38	2.180,10	2.136,57	2.095,58	2.056,90	2.020,33	1.877,90
7,68%	1,24	2.132,51	2.087,20	2.044,67	2.004,69	1.967,02	1.931,48	1.897,89	1.866,09	1.741,74
8,28%	1,34	1.972,46	1.932,76	1.895,44	1.860,30	1.827,14	1.795,81	1.766,16	1.738,06	1.627,83
8,90%	1,44	1.838,52	1.803,24	1.770,03	1.738,71	1.709,13	1.681,14	1.654,63	1.629,47	1.530,55
9,54%	1,54	1.724,24	1.692,52	1.662,63	1.634,41	1.607,72	1.582,45	1.558,49	1.535,73	1.446,07
10,20%	1,64	1.625,18	1.596,38	1.569,21	1.543,54	1.519,25	1.496,23	1.474,38	1.453,62	1.371,65
10,88%	1,74	1.538,14	1.511,78	1.486,90	1.463,37	1.441,09	1.419,95	1.399,88	1.380,79	1.305,32
12,25%	1,84	1.446,03	1.421,74	1.398,79	1.377,08	1.356,50	1.336,98	1.318,42	1.300,76	1.230,86
WACC	Beta									

Note: Valuation as of December 31, 2023.

Source: Own calculations

Figure 9 shows the results of changes in the risk-free rate, WACC, and beta. As of December 31, 2023, given the economic conditions of higher inflation and interest rates, the spot risk-free rate is relatively high compared to historical estimates, particularly from the past decade. In Figure 9 the share price of Rockwool is calculated for various risk-free rates with an interval of ± 0.1 percent point. If the current economic situation with elevated interest rates persists, one could argue that a potential future decline in the risk-free rate would result in a higher share price for Rockwool ceteris paribus, as observed in the figure.

Figure 9. Sensitivity analysis with regards to the risk-free rate, beta, and WACC

		Risk-Free Rate								
		1,63%	1,73%	1,83%	1,93%	2,03%	2,13%	2,23%	2,33%	2,43%
6,12%	0,94	2.916,10	2.832,84	2.755,77	2.684,20	2.617,58	2.555,41	2.497,26	2.442,75	2.391,55
7,11%	1,14	2.302,07	2.257,17	2.214,74	2.174,61	2.136,57	2.100,49	2.066,19	2.033,57	2.002,49
7,68%	1,24	2.097,18	2.062,18	2.028,90	1.997,22	1.967,02	1.938,20	1.910,67	1.884,34	1.859,14
8,28%	1,34	1.932,18	1.904,15	1.877,35	1.851,70	1.827,14	1.803,60	1.781,01	1.759,32	1.738,47
8,90%	1,44	1.795,68	1.772,72	1.750,67	1.729,49	1.709,13	1.689,53	1.670,67	1.652,48	1.634,95
9,54%	1,54	1.680,27	1.661,12	1.642,67	1.624,88	1.607,72	1.591,16	1.575,16	1.559,70	1.544,76
10,20%	1,64	1.580,95	1.564,73	1.549,06	1.533,91	1.519,25	1.505,07	1.491,34	1.478,03	1.465,14
10,88%	1,74	1.494,19	1.480,28	1.466,80	1.453,75	1.441,09	1.428,80	1.416,89	1.405,31	1.394,07
12,25%	1,84	1.402,69	1.390,63	1.378,92	1.367,55	1.356,50	1.345,77	1.335,32	1.325,17	1.315,28
WACC	Beta									

Note: Valuation as of December 31, 2023.

Source: Own calculations

Additionally, considering the terminal growth rate, an increase in this rate would lead to a higher share price due to the resulting increase in projected cash flows. Thus, a higher terminal growth rate implies greater future cash flows, which enhances the overall valuation of the company.

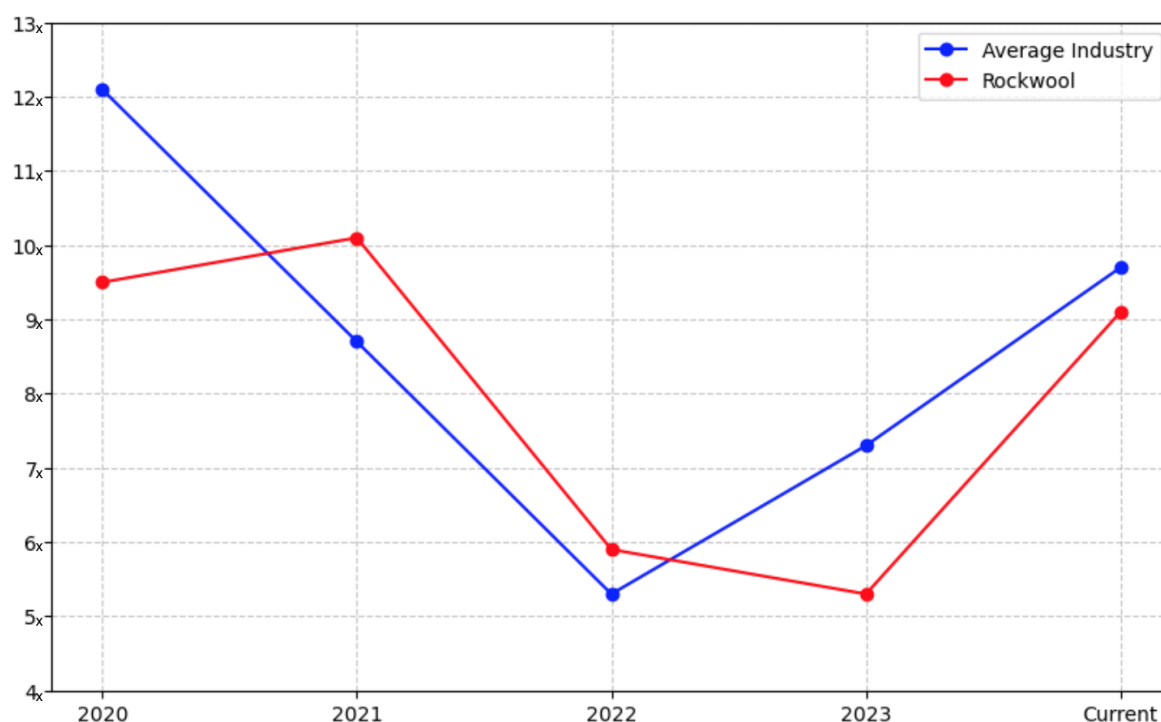
In conclusion, utilizing changes in the components of the WACC would significantly change the share price estimates with only minor changes in these components. Rockwool's net financial obligations further emphasizes the sensitivity of WACC to changes in beta. Considering a beta of 1.44 and the relatively high contemporary risk-free rate, combined with a relatively high market return as discussed in subsubsection 6.2.5, our estimated share price of DKK 1,709.13 is on the lower end.

8.3 Multiple Analysis

In this section, we perform a relative multiple analysis to evaluate our intrinsic DCF valuation using the EV/EBITDA multiple. EV/EBITDA is a widely used valuation metric that normalizes enterprise

value across different companies. This metric is particularly useful as it accounts for variations in capital structure, taxation, and depreciation policies, making it effective for comparing companies within the same industry. According to Penman (2013)⁵⁴, EV/EBITDA is particularly advantageous for evaluating firms with significant non-cash charges and varying tax environments, as it emphasizes operational profitability and cash generation capabilities. Close competitors to Rockwool, previously identified in this paper, are again selected as peers, as described in subsubsection A.2.1. First, we aim to examine the peers to determine if Rockwool A/S can be accurately evaluated within the context of its competitive industry, to do this we observe the evolution of the prior multiples.

Figure 10. Comparison of Average Industry and Rockwool A/S EV/EBITDA Multiples



Note: Average Industry contains firms from Table 12

Source: Own calculations

From Figure 10, we observe that Rockwool A/S's historical multiple has generally followed that of its peers. From 2021 to 2022, Rockwool's multiple was higher than that of the peer group, indicating signs of overvaluation compared to its peers. However, from 2022 to the present, we observe the opposite situation, where Rockwool's multiple is slightly below the peer group, suggesting signs of undervaluation. Nonetheless, Rockwool's multiple does follow the overall trend of the peer group, confirming the relevance of the multiple peer comparison.

⁵⁴Penman (2013) [35]

Table 14. Peers EV/EBITDA Multiples

Company	L3Y	2023	Current
Saint-Gobain	5.8x	5.8x	6.8x
Owens Corning	10.5x	6.4x	8.3x
Kingspan Group	17.0x	14.3x	16.4x
CSR Limited	6.8x	4.9x	12.1x
Patrick Industries, Inc	7.2x	8.5x	10.0x
STO SE & CO. KGAA	4.8x	3.8x	4.8x
Median	7.3x	6.1x	9.1x
Average	8.7x	7.3x	9.7x

Note: Based on company annual reports and own calculations

Source: Yahoo Finance

Due to a lack of data, we were able to calculate the multiple for the peer group only for the past three years. Considering the significant economic changes since the onset of the COVID-19 pandemic, we applied the calculated average of the peer group's multiple to evaluate Rockwool's price per share for the end of 2023, as done in our DCF valuation. We then compared the current multiple to the calculated current share price of 03/31/2023 to validate the relevance of the peer group comparison. This approach ensures that our valuation accounts for recent economic fluctuations and leverages peer group multiples for a more accurate assessment, as suggested by valuation theory Damodaran (2006).⁵⁵.

Table 15. EV/EBITDA Multiple Valuations

	2023	Current (Q1 2024)
Peer group multiple (average)	7.3x	9.7x
Share price (DKK)	1,933	2,860
Actual share price (DKK)	1,976	2,269

Source: Own calculations

Table 15 summarizes the EV/EBITDA multiple analyses and valuations as of December 31, 2023, and Q1 2024. This alignment, of the calculated prices and the actual prices confirms the validity of using EV/EBITDA multiples for valuing Rockwool and the reliability of the peer group. Furthermore the alignment suggests that the market's expectations are in line with the Rockwool's performance and outlook.

9 Conclusion

In this paper, we determined the intrinsic value of Rockwool A/S as of December 31, 2023, to be DKK 1,709.13 per share using a Discounted Cash Flow (DCF) model. This valuation falls slightly below the actual share price of DKK 1,976.00. Our valuation utilized a cost of capital of 8.90%, composed of a market risk premium of 8.65% and a cost of equity of 2.42%. The market risk premium was

⁵⁵Damodaran (2006)[19]

derived from Rockwool's beta value of 1.44 against the MSCI World index, a risk-free rate of 3.88% based on the US 10-year government bond, resulting in a implied market risk of 4.60%.

Despite a methodically calculated cost of capital, the valuation remains uncertain. We employed a 10-year forecast assuming a lower growth rate converging towards a terminal rate of 3.50%, below Rockwool's historical five-year growth. The market may anticipate a higher sustainable growth rate for Rockwool, potentially justifying a higher valuation. Additionally, the risk-free rate used was slightly above recent historical levels, and the beta was calculated against the broad MSCI World index, which may not align closely with Rockwool investor perspectives.

Our sensitivity analysis highlighted the impact of these metrics on valuation accuracy, suggesting that a different approach could align our estimate more closely with the market value. A secondary valuation using the EV/EBITDA multiples from industry peers yielded estimates very close to market valuations, supporting the efficient market hypothesis.

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A Appendix

A.1 PESTEL-analysis

A.1.1 Political

As for all global companies and manufacturers, the Russian invasion of Ukraine has a great impact of the political environment going forward. 18% of Rockwool's total revenue was in 2023 credited to the Eastern European and Russian market, meaning that political and economic sanctions made against Russia from the western world crucially affects the business operation of Rockwool. 10% of all Rockwool employees are located in Russia and the company continues to operate their 4 Russian factories. Their Russian operations has made the National Corruption Agency of Ukraine deem Rockwool as a war sponsor for Russia as their building and insulation material were found on Russian warships. The consequences of this are hard to grasp for the time being due to the constant chances in conflict but it definitely adds a level of uncertainty to their Russian operation as well as their reputation in the rest of the world. Rockwool are yet to experience any significant drops in sales in their western operation due to this but future escalation in the conflict might affect this. Actually, Rockwool's revenue in Eastern Europe and Russia has remained fairly stable through out the conflicts but as they continue to operate and manufacture in the Russian market, the risks of further damage of the company name and seizure of assets remain. However, Rockwool operates mainly in the B2B-sector in the Western world or other parts of the world that are associated with greater political stability, which brings down the risk of substantial political damage on their business operations.

A.1.2 Economic

As for the rest of the world, Rockwool faces a great challenge with the current global economic climates with high inflation. The first and immediate threat to the operation and profitability of Rockwool is rise in labor cost. Rockwool's 11.993 employees (as of 2023) contributes to a labor cost of 817 million euro which makes up for a substantial part of their operational costs. We note that the OECD predicts a fall in inflation for the 20 biggest economies in the world which lowers the risk of significant labor cost rise in comparison to the prices of their product.

The increased inflation poses another threat for a company like Rockwool. Rockwool are heavily reliant on a healthy and growing building market as their product are almost entirely related to

construction. A high level of inflation often means that the demand for urban construction and renovation falls. The entire real estate and construction market are strongly linked to macroeconomic dynamics such as inflation and GDP growth. A slump in GDP growth poses a risk of further decrease in demand for real estate which have to be considered when analyzing the future economic risks of Rockwool's operation. When looking at the key figures in the OECD-forecast there are however no alarm bell ringings as global level are predicted as stable in the upcoming years. The current macroeconomic climate poses as an uncertain element when valuating Rockwool but as the OECD forecast predicts a greater level of stability for the upcoming year it will not be considered as a significant risk when forecasting the financial results of Rockwool later in the paper.

Table 16: OECD's Real GDP and Inflation forecast of the G20-countries (2023-2033)

	2023	2024	2025	2026	2027	2028	2029	2030	2031	2032	2033
Real GDP Growth	3,16%	2,91%	3,08%	3,20%	3,09%	2,94%	2,81%	2,72%	2,66%	2,62%	2,58%
Global Inflation	5,03%	3,43%	2,00%	2,00%	2,00%	2,00%	2,00%	2,00%	2,00%	2,00%	2,00%

Sources: OECD [6].

A.1.3 Sociocultural

Rockwool's sociocultural landscape, as reflected in its 2023 sustainability report, is shaped by a strong commitment to diversity, equity, and inclusion. With employees from over 83 nationalities working in 40 countries, the company recognizes the value of a diverse workforce. In 2023, they formalized a Diversity, Equity, and Inclusion policy to ensure equitable access to opportunities and create a supportive work environment. Their ambition to reach 35% female representation in leadership roles by 2024 demonstrates their commitment to inclusivity.

In its manufacturing communities, Rockwool is committed to positively influencing local economies through job creation, partnerships, and sustainability. Their emphasis on health and safety is evident in their "zero fatalities and serious accidents" goal, which prioritizes employee well-being. They also align their products with high environmental and health standards, such as LEED and WELL, to create safer and more sustainable buildings.

In addition, Rockwool fully embraces the UN Sustainable Development Goals, focusing on reducing waste and emissions, improving water use efficiency, and ensuring the recyclability of their products. By designing their stone wool insulation for fire safety, energy efficiency, and recyclability, they offer urban solutions that address critical global challenges. For a financial viewpoint Rockwool meets the criteria for a sustainable company that should fear no fiscal backlash from a lack of attention to social and cultural trends - with an exception for their continued operation in Russia. which we touched upon in section 4.1.⁵⁶

A.1.4 Technological

In Rockwool's technological strategy, the company's focus is on innovations that align with their sustainability and financial goals. Their 2023 sustainability report emphasizes a significant investment in decarbonization through factory electrification, specifically in their melting processes. Electrifying this critical phase minimizes greenhouse gas emissions, driving progress towards the 2034 target of reducing direct emissions by 38%, ultimately leading to their 2050 net-zero emissions commitment.

Further technological advances include expanding the Rockcycle® program, which involves developing more sustainable collection bags with 30% recycled content, reducing trips needed for recycling

⁵⁶Sustainability report 2023[8]

stone wool. The program’s geographic reach grew to 21 countries, reducing environmental impact while promoting a circular economy.

Additionally, Rockwool’s products are engineered for durability, recyclability, and multifunctionality. Their stone wool forms the basis for innovative solutions in building insulation, acoustic ceilings, horticultural applications, and industrial use. This versatility allows their technological investments to directly address global challenges like energy efficiency, fire resilience, noise reduction, and water conservation. Rockwool’s main markets are very developed and the company remains front runner in the technological development which means there are no significant financial risk or premiums to consider when valuating the firm.⁵⁷

A.1.5 Environmental and Legal

In the environmental aspect of a PESTEL analysis, Rockwool’s strategic direction aligns with the European Climate Law’s objectives of achieving net zero greenhouse gas emissions by 2050. This legislation mandates a 55% reduction in emissions by 2030 compared to 1990 levels, while establishing climate neutrality goals that require comprehensive emission reductions and investments in greener technologies across all sectors. Rockwool’s environmental strategy is geared toward electrifying its melting processes, expanding the Rockcycle® recycling program, and enhancing its product efficiency to meet these EU targets.

Legally, the European Climate Law also creates a system for monitoring and reviewing progress every five years, with the European Commission empowered to issue recommendations to member states if their national energy and climate plans (NECPs) lack alignment with the EU’s targets. The law promotes fairness and solidarity among member states to ensure consistent action toward climate neutrality.

Rockwool is positioning itself well within this framework by developing roadmaps and strategies that address emission reductions and sustainability. Its commitments include reducing direct emissions by 38% by 2034 and achieving net zero by 2050. The company’s proactive response to the evolving regulatory landscape ensures it remains compliant with EU climate policy and maintains its leadership in sustainable building solutions. These requirements puts Rockwool equal to their competitors for the European market but can pose a risk when business expansion in the third world is on the cards as environmental focus is less of a priority. It will, however, not be considered as a huge hindrance and financial obstacle when the forecast for the next 10 years is made later in the paper.⁵⁸⁵⁹

A.2 Porter’s Five Forces

A.2.1 Rivalry Within the Industry

In the competitive insulation industry, Rockwool A/S faces significant rivalry, especially from major global firms such as Saint-Gobain, Owens Corning, Kingspan Group etc. These companies, alongside Rockwool, are extensive multinational building material manufacturers with diverse product offerings and a focus on technological innovations for enhanced sustainability and energy efficiency. Rockwool’s competitive edge is bolstered by its intellectual property portfolio, which includes numerous patents that secure its innovative stone wool insulation solutions⁶⁰. This patent advantage supports Rockwool’s market position by protecting its unique production processes and product enhancements, thus

⁵⁷Sustainability 2023[8]

⁵⁸Sustainability report 2023[8]

⁵⁹Rockwool website, About us[11]

⁶⁰Mordor Intelligence™ [30]

differentiating it from competitors. Coupled with a strong commitment to sustainability, Rockwool appeals to an expanding demographic of environmentally conscious consumers across its key markets in Europe, North America, and Asia. Therefore, rating the rivalry within the industry as 4 but for the current situation and the future because of the difficult entry barriers.

A.2.2 Threat of Substitute Products

For Rockwool A/S, the threat of substitute products primarily comes from alternative cheaper insulation materials such as fiberglass, foam, and cellulose. While these substitutes often offer lower-cost solutions, Rockwool's stone wool insulation is distinguished by fire resistance, durability, and acoustic properties. Therefore rating this category by 3. According to their 2023 annual report, Rockwool's focus on non-combustibility and sustainability aligns with increasing global standards for fire safety and energy efficiency, which mitigates the risk posed by substitutes. Moreover, the company's investments in sustainability and product innovation further strengthen its market position against these alternatives. Future research may result in cheaper and better quality substitute products ranking the threat of substitute products in the future as 4 in the radar chart.

A.2.3 Threat of New Entrants

The threat of new entrants in the insulation market faced by Rockwool A/S is moderated by several industry-specific barriers. Notably, Rockwool's strong financial position, characterized by a low debt level as highlighted in their 2023 annual report, allows for strategic flexibility and resilience against economic fluctuations. The insulation industry requires significant capital investment in manufacturing technology and compliance with stringent environmental regulations, which can hinder new companies. Additionally, Rockwool's established brand and extensive distribution network provide a competitive advantage that new entrants might find challenging to replicate. Despite the potential for high returns in this market, the high entry barriers related to capital requirements and regulatory compliance must strengthen Rockwool's position against the threat of new competitors, hereby rating this category as 3 in influence. New production methods may be discovered in the future, or substitute products may become more easily accessible, increasing the rate of influence to 3.5 in the future.

A.2.4 Bargaining Power of Buyers

Rockwool A/S operates in a market where buyers range from large construction companies to individual homeowners, each possessing varying levels of purchasing power. While larger buyers may exert considerable pressure on pricing and payment terms due to their significant purchase volumes, Rockwool's diverse product range and superior product attributes, such as fire resistance and sustainability, help mitigate buyer power. Additionally, Rockwool's strong market presence and reputation for quality across various geographic regions enable it to maintain a balanced negotiation stance with its customers. Thus we assess the bargaining power of buyers in the insulation industry, particularly for Rockwool A/S, to be moderate with a rating of 2 and in the future with potentially more rivals or substitute products to 3.

A.2.5 Bargaining Power of Suppliers

The bargaining power of suppliers in the insulation industry, specifically for Rockwool A/S, is moderated by Rockwool's strategic sourcing and diversified supplier base. Despite the challenges in the global supply chain, Rockwool has maintained a robust procurement strategy, as evidenced in their

2023 annual report. For example, despite its operations in geopolitically sensitive areas like Russia, Rockwool has effectively managed its supplier relationships to ensure steady material supplies without compromising its market position or financial performance. This is reflected in their strong financial outcomes, even in volatile markets. Rockwool's ability to leverage its size and global presence enables the company to negotiate favorable terms, reducing the overall supplier power with a rating of 2.5 and a future rating of 3.

A.3 Reformulation of financial statements

Table 17: Reformulation of Statement of Equity (2017-2023)

<i>mEUR</i>	2017	2018	2019	2020	2021	2022	2023
Balance 01.01	1.536	1.684	1.877	2.118	2.092	2.398	2.584
Profit for the year	214	265	285	251	303	273	389
Other comprehensive income	-20	-20	44	-107	104	13	-63
Coprehensive income for the year	194	245	329	144	407	286	326
Sale and purchase of own shares	6	20	-2	-	-	-	-
Share buy-back program	-	-	-	-77	-3	-	-
Purchase of own shares	-	-	-	-3	-4	-2	-3
Sale of own shares	-	-	-	2	1	1	-
Expensed value of RSUs issued	-	1	1	2	2	2	2
Transactions non-controlling interest	3	-3	-	-	-4	-	-
Dividend paid to the share holders	-55	-70	-87	-94	-93	-101	-101
Sum	148	193	241	-26	306	186	224
Balance 31.12	1.684	1.877	2.118	2.092	2.398	2.584	2.808

Sources: Annual report (2018-2023) and own calculation.

Table 18: Balance sheet (2018-2023)

<i>mEUR</i>	2023	2022	2021	2020	2019	2018
Goodwill	98	107	102	96	97	95
Software	10	14	9	13	13	13
Customer relationships	23	29	34	39	43	49
Other intangible assets	8	12	14	21	21	19
software in progress	6	3	17	12	18	13
Total intangible assets	145	165	176	181	192	189
Buildings and sites	881	908	860	637	643	608
Plant and machinery	740	702	680	439	444	444
Other operating equipment	26	18	16	22	20	15
Tangible assets in progress	432	359	273	534	399	160
Total tangible assets	2.079	1.987	1.829	1.632	1.506	1.227
Right-of-use assets	72	88	61	44	52	-
Shares in associated companies	11	9	8	6	6	5
Long-term deposits and receivables	8	4	3	10	15	1
Deferred tax assets	46	48	52	54	54	46
Total financial assets	65	61	63	70	75	52
Non-current assets	2.361	2.301	2.129	1.927	1.825	1.468
Inventories	375	433	317	216	236	238
Trade receivables	337	347	307	247	275	274
Other receivables	53	80	98	60	54	42
Prepayments	30	31	24	15	15	18
Income tax receivable	44	27	39	38	14	5
Cash	354	209	166	241	275	386
Current assets	1.193	1.127	951	817	869	963
Assets	3.554	3.428	3.080	2.744	2.694	2.431
<i>mEUR</i>	2023	2022	2021	2020	2019	2018
Share capital	29	29	29	29	29	29
Currency translation adjustment	-171	-117	-134	-212	-104	-157
Proposed dividend	125	102	102	94	94	88
Retained earnings	2.824	2.567	2.398	2.178	2.096	1.912
Hedging	-3	-1	-1	-1	-1	1
Non-controlling interest	-	-	-	4	4	4
Total Equity	2.804	2.580	2.394	2.092	2.118	1.877
Deferred tax liabilities	66	55	51	47	43	51
Pension obligations	39	32	35	66	62	53
Lease liabilities	48	61	44	27	34	-
Provisions	22	19	16	18	17	15
Bank loans and other loans	24	39	17	-	4	2
Non current liabilities	199	206	163	158	160	121
Short-term portion of bank loans and other loans	13	102	9	100	1	3
Bank debt	1	7	1	1	6	6
Trade payables	241	270	283	184	196	209
Lease liabilities	29	24	19	18	18	-
Provisions	15	17	8	8	9	7
Income tax payable	41	26	26	25	29	34
Other payables	211	196	177	158	157	174
Current liabilities	551	642	523	494	416	433
Total liabilities	750	848	686	652	576	554
Total equity and liabilities	3.554	3.428	3.080	2.744	2.694	2.431

Sources: Annual report (2018-2023) and own calculation.

Table 19: Reformulation of Balance sheet (2018-2023)

<i>mELFI</i>	2023	2022	2021	2020	2019	2018
Operating assets						
Goodwill	98	107	102	96	97	95
Software	10	14	9	13	13	13
Customer relationships	23	29	34	39	43	49
Other intangible assets	8	12	14	21	21	19
Software in progress	6	3	17	12	18	13
Buildings and sites	881	908	860	637	643	608
Plant and machinery	740	702	680	439	444	444
Other operating equipment	26	18	16	22	20	15
Tangible assets in progress	432	359	273	534	399	160
Right of use assets	72	88	61	44	52	-
Deferred tax assets	46	48	52	54	54	46
Long-term deposits and receivables	8	4	3	10	15	1
Inventories	375	433	317	216	236	238
Trade receivables	337	347	307	247	275	274
Other receivables	53	80	98	60	54	42
Prepayments	30	31	24	15	15	18
Income tax receivable	44	27	39	38	14	5
Total Operating assets	3.189	3.210	2.906	2.497	2.413	2.040
Operating liabilities						
<i>Non - Current</i>						
Deferred tax liabilities	66	55	51	47	43	51
Provisions	22	19	16	18	17	15
<i>Current</i>						
Trade payables	241	270	283	184	196	209
Provisions	15	17	8	8	9	7
Income tax payable	41	26	26	25	29	34
Other payables	211	196	177	158	157	174
Total operating liabilities	596	583	561	440	451	490
Net operating assets (NOA)	2.593	2.627	2.345	2.057	1.962	1.550
<i>mELFI</i>	2023	2022	2021	2020	2019	2018
Financial assets						
<i>Non - Current</i>						
Shares in associated companies	11	9	8	6	6	5
<i>Current</i>						
Cash	354	209	166	241	275	386
Total financial assets	365	218	174	247	281	391
Financial liabilities						
<i>Non - Current</i>						
Pension obligations	39	32	35	66	62	53
Lease liabilities	48	61	44	27	34	-
Bank loans and other loans	24	39	17	-	4	2
<i>Current</i>						
Short-term portion of bank loans and other loans	13	102	9	100	1	3
Bank debt	1	7	1	1	6	6
Lease liabilities	29	24	19	18	18	-
Total financial liabilities	154	265	125	212	125	64
Net financial obligations (NFO) (Net Financial Assets (NFA))	-211	47	-49	-35	-156	-327
Minority interest (MIN)	0	0	0	4	4	4
Common shareholders equity (CSE)	2.804	2.580	2.394	2.088	2.114	1.873

Sources: Annual report (2018-2023) and own calculation.

Table 20: Statement of comprehensive income (2018-2023)

<i>mELFI</i>	2023	2022	2021	2020	2019	2018
Net sales	3.620	3.907	3.088	2.602	2.757	2.671
Other operating income	15	15	6	6	16	11
Operating income	3.635	3.922	3.094	2.608	2.773	2.682
Raw material costs and production material costs	1.302	1.682	1.116	845	929	909
Delivery costs and indirect costs	437	518	438	363	381	385
Other external costs	300	273	205	184	218	232
Personnel costs	817	811	733	694	697	649
Operating costs	2.856	3.284	2.492	2.086	2.225	2.175
EBITDA	779	638	602	522	548	507
Amortisation, depreciation and impairment	261	236	201	184	176	166
EBIT	518	402	401	338	372	341
Income from investments in associated companies	2	1	1	1	-	1
Financial income	38	16	10	8	5	5
Financial expenses	-36	-61	-19	-22	-10	-12
Profit before tax	522	358	393	325	367	335
Tax on profit for the year	133	85	90	74	82	70
Profit for the year	389	273	303	251	285	265
Other comprehensive income	-63	13	104	-107	44	-20
Comprehensive income for the year	326	286	407	144	329	245

Sources: Annual report (2018-2023) and own calculation.

Table 21: Reformulation of statement of comprehensive income (2018-2023)

<i>mEUR</i>	2023	2022	2021	2020	2019	2018
Revenue	3.635	3.922	3.094	2.608	2.773	2.682
Cost of sales	-1.302	-1.682	-1.116	-845	-929	-909
Gross profit	2.333	2.240	1.978	1.763	1.844	1.773
Delivery costs and indirect costs	-437	-518	-438	-363	-381	-385
Other external costs	-300	-273	-205	-184	-218	-232
Personnel costs	-817	-811	-733	-694	-697	-649
EBIDTA	779	638	602	522	548	507
Amortisation, depreciation and impairment	-261	-236	-201	-184	-176	-166
Core operating profit from sales, before tax (Core EBIT)	518	402	401	338	372	341
Income tax expense	-133	-85	-90	-74	-82	-70
Tax benefit on debt	1	-10	-2	-3	-1	-1
Tax on operating profit from sales	-132	-95	-92	-77	-83	-71
Core operating profit, after tax	386	307	309	261	289	270
Actuarial gains and losses of pensions obligations	-10	-	28	-3	-10	1
Currency adjustment from translation of entities	-54	17	78	-108	53	-22
Hedging instruments, value adjustments	-3	1	-	2	-3	-
Tax on other comprehensive income	4	-5	-2	2	4	1
Total other comprehensive income	-63	13	104	-107	44	-20
Total operating profit, after tax	323	320	413	154	333	250
Financial income	40	17	11	9	5	6
Financial expenses	-36	-61	-19	-22	-10	-12
Net financial cost (before tax)	4	-44	-8	-13	-5	-6
Tax benefit or payments	-1	10	2	3	1	1
Net financial cost (after tax)	3	-34	-6	-10	-4	-5
Total comprehensive income	326	286	407	144	329	245
<i>Reported comprehensive income</i>	326	286	407	144	329	245

Sources: Annual report (2018-2023) and own calculation.

Table 22: Statement of cash flows (2018-2023)

<i>mELUF</i>	2023	2022	2021	2020	2019	2018
EBIT	518	402	401	338	372	341
Adjustments for depreciation, amortisation and write-downs	261	236	201	184	176	166
Adjustments of non-cash operating items	-2	4	-5	7	6	-7
Changes in net working capital	71	-148	-82	27	-39	-22
Cash flow from operations before financial items and tax	848	494	515	556	515	478
Finance income etc. received	14	10	10	8	5	5
Finance costs etc. paid	-34	-38	-17	-22	-5	-12
Taxes paid	-121	-72	-82	-104	-113	-63
Cash flow from operating activities (CFFO)	707	394	426	438	402	408
Purchase of tangible assets	-321	-326	-295	-358	-379	-204
Received investment grants	9	1	-	19	-	1
Purchase of intangible assets	-5	-8	-7	-4	-21	-19
Proceeds from sale of listed equities	-	-1	-	-	-	18
Business acquisitions, net of cash	5	-	-8	-19	-	-8
Cash flow from investing activities (CFFI)	-312	-334	-310	-362	-400	-212
Cash flow from operating and investing activities (free cash flow)	395	60	116	76	2	196
Dividend paid	-101	-102	-93	-94	-87	-70
Dividend paid to non-controlling interests	-	-	-3	-77	-	-
Payments for own shares	-3	-2	-4	-3	-2	-3
Proceeds from sale of own shares	-	1	1	2	-	23
Transactions with non-controlling interests	-	-	-8	-	-	-3
Installments of lease liabilities	-29	-26	-21	-20	-17	-
Increase in non-current receivables	-5	-	8	4	-14	-
Proceeds from borrowings	1	124	26	152	4	-
Installments of non-current debt	-101	-9	-100	-56	-4	-1
Cash flow from financing activities (CFFF)	-238	-14	-194	-92	-120	-54
Net Cash Flow	157	46	-78	-16	-118	142
Cash available 1/1 primo	202	165	240	269	380	243
Exchange rate adjustments	-6	-9	3	-13	7	-5
Cash available 31/12 ultimo	353	202	165	240	269	380

Sources: Annual report (2018-2023) and own calculation.

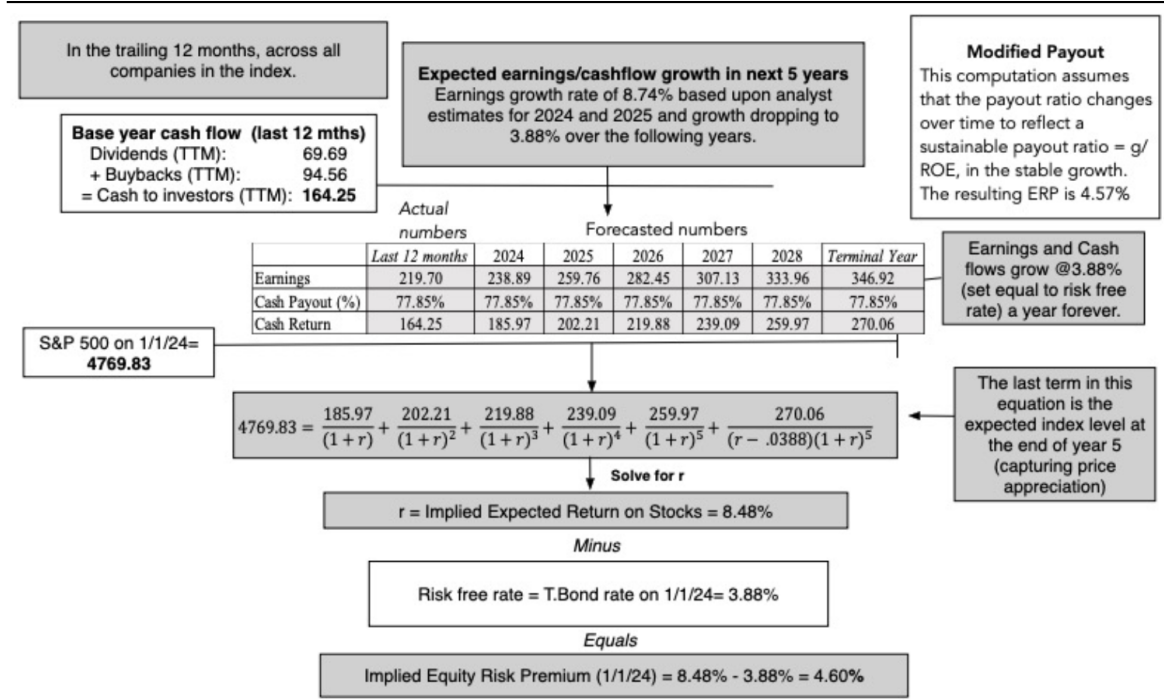
Table 23: Free Cash Flow calculations (2019-2023)

	2023	2022	2021	2020	2019
Method 1: FCF = OI - ΔNOA					
OI	323	320	413	154	333
ΔNOA	-34	282	288	95	412
FCF	357	38	125	59	-79
Method 2: FCF = NFE - ΔNFO + d + MININC - ΔMIN					
NFE	-3	34	6	10	4
ΔNFO	-258	96	-14	121	171
d	102	100	101	170	88
MININC	0	0	0	0	0
ΔMIN	0	0	-4	0	0
FCF	357	38	125	59	-79

Sources: Annual report (2018-2023) and own calculation.

A.4 Implied Market Risk Premium

Figure 11. Damodaran's Calculation of MRP



Source: Damodaran (2024)[23]

A.5 Key formulas in the decomposition of the ROCE

$$ROCE_t = \frac{CI_t}{\frac{1}{2}(CSE_t + CSE_{t-1})} = RNOA_t + (FLEV_t \cdot SPREAD_t) \quad (4)$$

$$RNOA_t = \frac{OI \text{ after tax}_t}{\frac{1}{2}(NOA_t + NOA_{t-1})} = PM_t \cdot ATO_t \quad (5)$$

$$FLEV_t = \frac{NFO_t + NFO_{t-1}}{CSE_t + CSE_{t-1}} \quad (6)$$

$$SPREAD_t = RNOA_t - NBC_t \quad (7)$$

$$NBC_t = \frac{NFE_t}{\frac{1}{2}(NFO_t + NFO_{t-1})} \quad (8)$$

$$PM_t = \frac{OI \text{ after tax}_t}{Revenue_t} \quad (9)$$

$$ATO_t = \frac{Revenue_t}{\frac{1}{2}(NOA_t + NOA_{t-1})} \quad (10)$$

A.6 Historical Market Risk

A.6.1 Time period

Determining the appropriate time period to estimate historical risk premiums is crucial, as it can significantly affect the results. Analysts who prefer using data from the inception date argue that it provides a more comprehensive view. However, many analysts advocate for using shorter periods, such as fifty, twenty, or even ten years, suggesting that these periods offer a more updated estimate that reflects current market conditions and investor risk aversion. Using shorter periods can provide a more contemporary view of the market but introduces greater noise in the risk premium estimate. the standard error for a 10-year estimation period can be as high as 6.33%⁶¹, which is comparable to or even larger than the actual estimated premium. This significant noise can outweigh the benefits of using more recent data. However, risk premiums from calculated on longer periods may not be relevant for today's market conditions due to lack of reliability. Thus, choosing the right time period for estimating the historical market risk premium involves balancing the need for relevance and the desire to minimize statistical noise. Different time periods are therefore included in our analysis of the market risk premium.

A.6.2 Market Index

The historical risk premium is influenced by how stock returns are estimated. While using an index with a long history, such as the Dow 30, might seem ideal, it may not accurately reflect overall stock returns. Ideally, the broadest index of stocks should be used to compute returns, considering two essential factors. The index must be market-weighted because the overall returns on equities will be skewed towards larger market cap stocks. Rockwool A/S is included in the C25 stock index therefore this would be ideal to include for the market risk premium. While it is useful for understanding the Danish equity market, it has limitations when used for calculating the market risk premium for a global company like Rockwool A/S. The C25 Index includes only 25 companies, which is a relatively small sample size. This limited scope can lead to a lack of diversification and may not accurately reflect the broader market dynamics that a company like Rockwool might encounter.

For our analysis, we will therefore use the MSCI World Index and the S&P500 to calculate the market risk premium. The MSCI World Index is a broad, market-weighted index that includes large and mid-cap stocks across 23 developed markets, providing a comprehensive view of global equity returns. The S&P 500, while limited to the U.S., is also market-weighted and includes a diverse range of large-cap stocks, making it a reliable indicator of global market performance.

A.6.3 Measures of Market Risk Premium

Estimating the market risk premium accurately is crucial for financial analysis, involving several methodologies, each with its strengths and limitations. This section discusses the arithmetic average return and the geometric average return.

The arithmetic average return is the simple mean of periodic returns, calculated by summing the returns and dividing by the number of periods. This method is useful for estimating expected returns over a single period. According to Damodaran (2024)⁶², the arithmetic average is typically higher than the geometric average because it does not account for compounding effects. It is most appropriate for short-term horizons where returns are assumed to be independent of each other.

⁶¹ $20\%/\sqrt{10} = 6.33\%$ [24]

⁶²Damodaran (2024) [21]

The geometric average return accounts for the compounding effect of returns over multiple periods. Damodaran (2024)⁶³ notes that the geometric average provides a lower estimate than the arithmetic average because it smooths out volatility, making it suitable for long-term investment horizons. In conclusion, we choose to use the geometric mean for the valuation of Rockwool A/S because it provides a more accurate reflection of long-term growth by accounting for the compounding effect of returns and smoothing out volatility, making it the most reliable measure for long-term financial projections. This approach is supported by academic research, such as Mehra and Prescott (1985)⁶⁴, which highlights the importance of considering compounded growth rates in long-term return estimates. Underneath in Table 24 the calculated means are shown respectively to their time period.

Table 24: Historical Annualized Market Return

Years	S&P 500			MSCI World Index		
	30	40	50	30	40	50
<i>Standard Error</i>	3.65%	3.16%	2.83%	3.65%	3.16%	2.83%
Arithmetic Mean	9.67%	10.13%	9.52%	9.71%	10.56%	10.84%
Geometric Mean	8.03%	8.97%	7.52%	8.10%	8.86%	9.23%

Source: Yahoo Finance

Furthermore, according to Damodaran (2024)⁶⁵, using longer time periods, such as 50 years, can provide a more stable and reliable estimate of the market risk premium by reducing the impact of short-term fluctuations. Therefore, for our analysis, we will use the geometric mean for the MSCI World Index over a 50-year period, ensuring a robust and realistic estimate of the market risk premium that aligns with long-term investment horizons and financial valuation principles. Subtracting the risk-free rate of 3.88% according to the 10-year US government bond we obtain a market risk premium of 5.35% based on the historical returns.

In conclusion, we choose to apply the implied market risk premium over the survey and historical market risk premiums due to its relevance to real-time market data and objective measures, which provide a more accurate and current reflection of market conditions taking the idea of S&P500 as a broader market into consideration. This method reduces subjective biases in survey-based estimates and reduces the noise and potential inaccuracy associated with historical averages, ensuring a robust and reliable basis for our valuation of Rockwool A/S which aligns with the concepts of Damodaran (2024)⁶⁶.

⁶³Damodaran (2024) [21]

⁶⁴Mehra and Prescott (1985) [34]

⁶⁵Damodaran (2024)[22]

⁶⁶Damodaran (2024)[22]